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# Personalised telehealth intervention for chronic disease management: A pilot randomised controlled trial

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**Abstract**

**Introduction:** The aim of this study was to assess the impact of home-based telehealth monitoring on health outcomes, quality of life and costs over 12 months for patients with diabetes and/or chronic obstructive pulmonary disease (COPD) who were identified as being at high risk of readmission to hospital.

**Methods:** This pilot study was a randomised controlled trial combined with an economic analysis to examine the outcomes of standard care versus home-based telehealth for people with diabetes and/or COPD who were at risk of hospital readmission within one year. The primary outcomes were (i) hospital admission and length of stay (LOS); and (ii) health-related quality of life (HRQOL); and the secondary outcomes were (i) health-related clinical outcomes; (ii) anxiety and depression scores; and (iii) health literacy. The costs of the intervention and hospitalisations were included.

**Results:** A total of 86 and 85 participants were randomised to the intervention and control groups respectively. The difference between groups in hospital LOS was –3.89 (95% confidence interval (CI): –9.40, 1.62) days, and for HRQOL, 0.09 (95% CI: 0.05, 0.14) in favour of the telehealth monitoring group. There was a saving of AUD$6553 (95% CI: –12145, –961) in the cost of hospitalisation over 12 months, which offset the increased cost of tele-monitoring. The intervention group showed an improvement in anxiety, depression and health literacy at 12 months, and in the diabetes group, a reduction in microalbuminuria.

**Discussion:** The telehealth monitoring intervention improved patient's health outcomes and quality of life at no additional cost.

**Keywords**

Cost analysis, cost-effectiveness, home telecare, telecare, telehealth

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## Introduction

The burden and morbidity of chronic illness and aging in our community continues to place demands on all aspects of the health system in Australia and worldwide. This burden has been most notable in the increasing demand for hospital care, particularly with acute presentations of chronic conditions. Health funders provide a number of hospital avoidance and hospital substitution programs in Victoria, Australia. Despite these programs, and with a concerted effort towards improvement, many health services continue to struggle to improve efficiency in service delivery to meet demand. A recent systematic review reported that hospital-based programs do reduce chronic-disease-related hospital readmissions using interventions such as patient discharge planning, case management, telephone and scheduled follow-ups, tele-monitoring and patient education.[1] Evidence has also shown that patient transition from hospital to home depends on patients' capacity for self-care.[1] Other than physical health, psychological disturbances like anxiety and depression are prevalent in chronic diseases[2] and are related to excess morbidity and mortality. One review states that there is a very strong association between long-term physical illness and depression and anxiety.[2]

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The utilisation of technology has advanced in the general community; however, healthcare has been slow to adopt technologies that can be used by people in their home environment to improve their health-related quality of life (HRQOL).[3] Many evaluations of telehealth services lack the rigour to provide a succinct answer to the clinical effectiveness and cost-effectiveness of telehealth interventions (with consideration towards publication bias and short phases of delivery).[4] It has also been difficult to extrapolate data across different chronic diseases. Celler et al.[5] have shown that with time and improved technology the costs of telehealth applications are decreasing. This, coupled with advancement in telecommunications, holds promise for the future success of telehealth interventions. There is growing evidence of positive outcomes for patient's health and well-being, as well as for the efficient use of healthcare resources.[3]

A challenge to the successful implementation of chronic disease interventions is to identify the patient population who are likely to benefit most in terms of a reduction in hospitalisations and/or an improvement in health outcomes. Cost-effectiveness is more likely to be demonstrated in a patient population that is at high risk of hospital readmission. The Barwon Health personalised telehealth remote patient monitoring intervention was developed to reduce hospitalisations in a population of patients with diabetes and/or chronic obstructive pulmonary disease (COPD) at risk of readmission. Identifying this population was, therefore, considered to be paramount to the success of the intervention in terms of outcomes and a reduction in the costs of hospitalisation. Using Pattern Recognition and Data Analytics (PRaDA),[6] hospitalisation variables such as diagnosis, previous admissions to hospital, co-morbidities, procedures and age were modelled to identify a cohort of patients with a likelihood of readmission to hospital within one year.

The objective of this study was to determine if a personalised telehealth remote patient monitoring intervention was more effective at no additional cost compared to usual care for people with diabetes and/or COPD who were at high risk of hospital readmission. The perspective of the study was that of the health service provider and the primary outcome was HRQOL.

## Methods

### Study design

A pilot single-blinded randomised controlled trial accompanied by an economic evaluation was undertaken to examine the clinical effectiveness and cost-effectiveness of a personalised telehealth remote patient monitoring intervention compared to usual care provided by Barwon Health and other community-based services. The study was approved by the Human Research Ethics Committees of Barwon Health (HREC 13/14) and Deakin University (HREC 2015-026). The trial was registered on the Australian New Zealand Clinical Trials Registry (ANZCTR) under ACTRN12617000396325.

### Participants and recruitment

People with diabetes and/or COPD who were admitted to Barwon Health for any reason between November 2012 and November 2013 were invited to participate in the trial if they (i) had a high likelihood of hospital readmission over the next 12 months of 0.5 to 0.8 according to the PRaDA model[6]; (ii) were able to enter and submit their own data via a computer or similar device; (iii) were English speaking; (iv) lived in their own home; and (v) were willing and able to take their own vital signs and biometric readings via an appropriate device. Participants were excluded if they had (i) high expected mortality over the next 12 months; (ii) pregnancy; (iii) vision or hearing impairment that impacted their ability to use the telehealth device or complete biometric readings; (iv) severe cognitive impairment as determined by clinical assessment; (v) sub-optimal management of mental illness as per clinical assessment; (vi) current enrolment in the Hospital Admission Reduction Program; (vii) active palliative care; or (viii) drug or alcohol dependency. Participants who responded to the invitation were offered a meeting in their own home with a study team member to learn about the trial. People who provided consent underwent a comprehensive assessment prior to randomisation. Those invited but who did not respond were followed-up by phone call on two occasions and, if there was no response, were not randomised to either intervention or control groups. All participants were recruited between December 2013 and June 2014.

### Randomisation

Participants who met inclusion criteria and agreed to participate were randomised to the intervention or control groups using sequential envelopes. There were separate envelopes for participants with diabetes, COPD and both diabetes and COPD. Baseline measurements for intervention and control participants were assessed at a second home visit. Following this second home visit, and after randomisation, the project manager, nursing teams and participants were aware of allocation to the intervention or control groups. If during the assessment phase of the pilot study a patient was identified as requiring a community-based service, these referrals were made regardless of the patient being a part of the intervention or control group.

### Intervention

The intervention occurred during the period from December 2013 to June 2015. All patients were expected to participate for 12 months, and those randomised to the intervention group were provided with an android tablet and peripheral equipment – for example, a sphygmomanometer – to enable them to collect biometric readings and enter daily answers to health questionnaires. A technology platform called Remote Patient Monitoring from TELUS® Canada[7] was used to facilitate the remote

patient monitoring service. Participants were provided with an individualised care plan, designed by the specialist consultant and remote patient monitoring nurse, to guide specifically designed management protocols tailored to their condition. Following patient assessment, the monitoring plan was completed with clinical parameters (alert thresholds) adjusted to the individual and approved by the treating medical consultant. These individual parameters included upper and lower limits to blood pressure, pulse rate, blood glucose readings and oxygen saturation levels.

The remote monitoring service allowed incoming data to be identified as being outside of usual parameters for that patient. The Barwon Health personalised telehealth monitoring program was staffed during office hours seven days a week by registered nurses with skills in the management of diabetes, respiratory conditions and community nursing. Outside of these hours, an SMS would be sent to on-call staff if an alert was raised. The nurse, in response to the alert, then made contact with the patient to support self-management strategies and to meet the clinical need arising for that patient. In addition to remote monitoring, participants received fortnightly health coaching sessions, usually via a videoconferencing facility. These coaching sessions allowed further planning and work with the participant on their individual goals and any other supports that they needed.

All participants were directly supported in the community by their general practitioner (GP) in meeting their overall medical needs. When required, participants were referred to their GP for medical care by the registered nurse. Communication between the nurse and GP would usually occur via written correspondence supplemented with phone calls for care planning if required. The pilot project did not provide an immediate response of a medical nature outside of usual GP care. Participants whose health decline was identified as acute in nature were referred to the emergency department or their GP. Participants were advised in writing and verbally that the telehealth service was not an emergency medical service and that they should seek emergency medical care if they believed it was appropriate.

Lead medical consultants (respiratory and diabetes) for each patient cohort were chosen from senior medical staff of University Hospital Geelong. The role of the lead medical officers was to oversee the plan of care for each patient and authorise individualised modified parameters for the monitoring system. Where complexity arose in the care plan for the treatment of a patient, the lead medical officers were involved in managing patient care as required. This, at times, included outpatient assessments in clinic.

### Control

Participants randomised to the control group were advised to manage their condition as usual, in conjunction with their community GP, Barwon Health outpatient clinic appointments and emergency department where appropriate.

### Outcomes

The primary outcomes were as follows (Table 1): (i) hospital admissions, length of stay (LOS) and costs of care (intervention and hospitalisation); and (ii) HRQOL. The secondary outcomes (Table 1) were as follows: (i) health-related clinical outcomes; (ii) anxiety and depression scores; and (iii) health literacy. Trained assessors (nursing staff) conducted face-to-face assessments in the participant's home at baseline and 12 months. At the initial assessment both the nurse and patient were not aware of the allocation to control or intervention groups. At the final assessment both nurse and client were aware of the study allocation. Data for HRQOL, clinical outcomes, anxiety and depression, and health literacy were collected at baseline and follow-up at 12 months. Data relating to hospitalisation were collected from Barwon Health at the end of the 12-month period (Table 1).

### Statistical analysis

Data were stored in password-protected files on computers at Barwon Health. All analyses were undertaken using Intercooled STATA®, version 14.2.[8] The baseline population characteristics were described using mean, standard deviation and *p*-values. Hospital admissions, LOS and cost data over 12 months were analysed by comparing study group means using a *t*-test and 95% confidence intervals (CIs). Analysis of covariance (ANCOVA) was used to compare the mean difference (MD) in health outcomes at baseline and 12 months between intervention and control groups.

During the course of the pilot study, a Data Safety Management Board was set up to monitor aggregated data on a monthly basis. In-depth reviews were undertaken of any serious adverse events, identified as admission to Intensive Care Unit, arrest and death of participants.

### Cost-effectiveness

The fixed and variable costs of running the intervention and the costs of hospitalisation from a health service perspective (Barwon Health) over a 12-month period were included in the economic evaluation. All costs were reported as 2014 Australian dollars. The mean cost difference between the two groups was determined using an independent *t*-test and 95% CIs to report statistical significance.

An incremental cost-effectiveness ratio (ICER) was calculated based on the difference in costs and HRQOL between baseline and 12 months derived from ANCOVA using the Assessment of Quality of Life (AQoL-8D) instrument.[9] Intention to treat analyses was undertaken based on group allocation for hospital costs and admission data, but not quality of life due to participant mortality ($n = 8$).

**Table 1.** Study outcomes.

| | Measures | Baseline | 12 months |
|---|---|---|---|
| Primary outcomes | | | |
| Hospital admissions, LOS and cost | Hospitalisation data over a 12-month period were collected from the hospital admission and costing system | – | * |
| Health-related quality of life (HRQOL) | HRQOL data were derived from the AQoL-8D utility instrument.[9] The AQoL-8D attributes a score to each of eight domains (independent living, happiness, mental health, coping, relationships, self-worth, pain, senses). These were converted to a health-related utility score based on Australian weights provided with the AQoL-8D[9] to derive a single, overall utility-based HRQOL measure | * | * |
| Secondary outcomes | | | |
| Health-related clinical outcomes | Change in 6-min walk test, COPD Assessment Test (CAT), Forced Expiratory Volume (FEV1), Forced Vital Capacity (FVC), Shortness of breath score (BORG), Oxygen saturation level ($SaO_2$), HbA1C, urine microalbumin and serum lipids | * | * |
| Health literacy | The Health Education Impact Questionnaire (HeiQ) assessed patient's health literacy[21] across eight domains: health-directed behaviour; positive and active engagement in life; self-monitoring and insight; constructive attitudes and approaches; skill and technique acquisition; social integration and support; health services navigation; and emotional distress | * | * |
| Anxiety and depression | The Generalised Anxiety and Depression Scale (GAD-7), 7 items assess patient's anxiety and depression level.[22] The total GAD-7 score ranges from 0 to 21 and is categorised according to severity. Minimal anxiety (score 1–4), mild anxiety (score 5–9), moderate anxiety (score 10–14) and severe anxiety (score > 15).<br>Depression level was assessed using Patient Health Questionnaire 9-item scale (PHQ-9),[23] 9 items assed general health. The total PHQ-9 score ranges from 0 to 27, with each of the 9 items scored from 0 to 3.[23] PHQ-9 categorised as minimal depression (score 5–9), mild depression (score 10–14), moderate depression (score 15–19) and severe depression (score > 20).[24] | * | * |

LOS: length of stay; AQoL-8D: Assessment of Quality of Life; COPD: chronic obstructive pulmonary disease.
*Data was collected at this time point.

Where there were missing data from loss to follow-up for the primary outcome, HRQOL, the imputation method used was the baseline value carried forward, or where there were missing values at baseline and 12 months, the group mean. The same method for missing values was used for secondary outcome data, Generalized Anxiety Disorder 7-item (GAD-7) and Patient Health Questionnaire 9-item (PHQ-9) scales, but not for clinical outcome data.

## Results

### *Study population*

There were 1605 patients identified as having a previous admission for either diabetes or COPD who had a high probability of hospital readmission in the next 12 months based on the PRaDA model. Of these, 711 people (50% of excluded patients) declined to participate in the study, 370 (26%) either did not meet the study inclusion criteria (diagnosis) or withdrew prior to the study commencing and 347 (24%) did not respond to the invitation (includes deceased). The remaining eligible population ($n = 177$) were randomised (Figure 1) to the intervention group ($n = 86$) with remote patient monitoring or the control group ($n = 85$) receiving usual care (Table 2). These numbers included 33 people who had a PRaDA score between 0.3 and 0.49 who were invited to attend and agreed to participate prior to randomisation to either intervention ($n = 12$) or control groups ($n = 21$). There was a difference in the PRaDA score between groups with the intervention group having a higher probability of readmission (0.56 versus 0.50; $p < 0.05$).

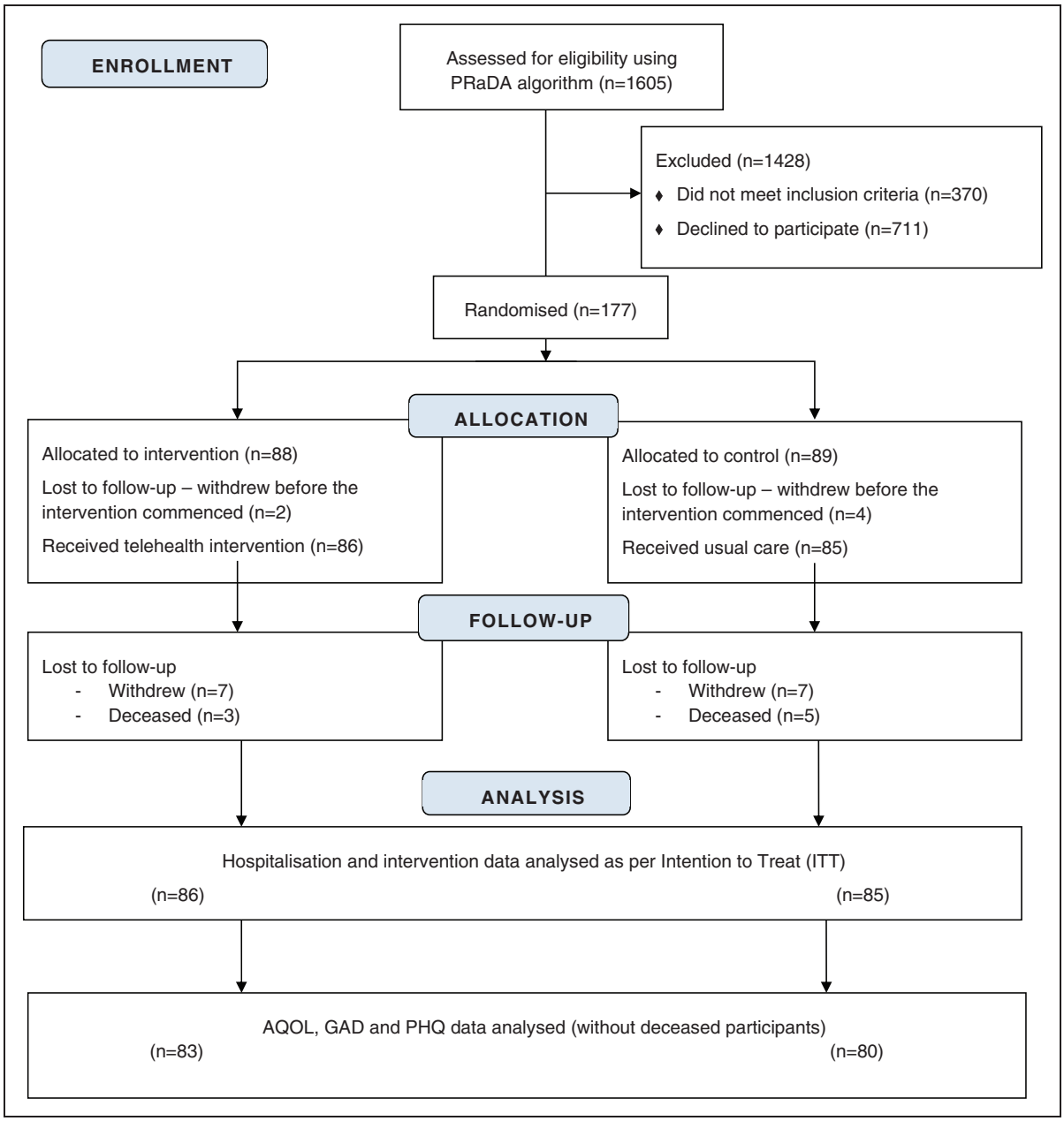


**Figure 1.** Consort chart.

*Primary outcomes*

1. Hospital admissions, length of stay and costs

There were a total of 232 admissions during the study period. Although the MD in the number of hospital admissions between the two groups was not statistically significant at 95%, there was a statistically significant difference in mean acute hospital LOS over 12 months (4.6 versus 8.7 days; 95% CI: −8.6, −0.4) and the mean cost of acute hospitalisation (AUD$4651 versus AUD$11,271; MD AUD$6553; 95% CI: −11528, −1712) in favour of the intervention group. The median costs of hospitalisation over 12 months were AUD$427 for the intervention group and AUD$847 for the control group.

2. Quality of life

There were 16 (19%) people in the intervention group and 25 (29%) people in the control group who had missing values for the AQoL-8D at 12 months. There was no difference in the mean AQoL-8D utility score between study groups at baseline. At 12 months the HRQOL utility score in the intervention group had increased by 0.05, and in the control group it had decreased by 0.04.

**Table 2.** Population demographic data (baseline).

| | Normal values | Intervention | | Control | | |
|---|---|---|---|---|---|---|
| | | *n* | Mean ± SD | *n* | Mean ± SD | *p*-value |
| Study participants | | 86 | | 85 | | 0.596 |
| Male | | 49 (60%) | | 40 (47%) | | |
| Female | | 37 (40%) | | 45 (53%) | | |
| Diagnosis | | | | | | 0.822 |
| Diabetes | | 59 (67%) | | 62 (73%) | | |
| COPD | | 19 (22%) | | 16 (19%) | | |
| Both | | 8 (11%) | | 7 (8%) | | |
| Age | | 86 | 70.70 ± 11.56 | 85 | 70.13 ± 13.26 | 0.383 |
| PRaDA score | | 86 | 0.56 ± 0.18 | 85 | 0.50 ± 0.21 | 0.033 |
| Blood pressure | | | | | | |
| Diastolic | <80 mmHg | 80 | 80.08 ± 19.97 | 73 | 84.03 ± 25.47 | 0.143 |
| Systolic | <120 mmHg | | 134.09 ± 23.25 | | 132.75 ± 24.71 | 0.356 |
| HbA1c** | 6.5–7% | 63 | 6.04 ± 1.74<br>5.74* | 56 | 5.37 ± 1.56<br>5.36* | 0.015 |
| Total cholesterol | <5.5 mmol/l | 62 | 3.91 ± 1.29 | 56 | 4.01 ± 1.12 | 0.343 |
| HDL cholesterol | 0.9–2 mmol/l | 62 | 1.13 ± 0.63 | 55 | 1.17 ± 0.37 | 0.351 |
| LDL cholesterol | 2–3.4 mmol/l | 51 | 1.87 ± 1.13 | 46 | 1.83 ± 0.924 | 0.424 |
| Triglyceride | <1.7 mmol/l | 62 | 2.07 ± 1.06 | 56 | 2.14 ± 0.95 | 0.343 |
| Microalbumin | <2.5 mg/mmol | 55 | 28.72 ± 74.68 | 45 | 12.73 ± 20.04 | 0.083 |
| Percent*** | normal | 62% | | 47% | | |
| 6-min walk test distance | 348–371 m[25] | 20 | 222.16 ± 115.93 | 16 | 213.09 ± 94.48 | 0.401 |
| Borg scale (dyspnoea) | Modified Borg scale: 0 (no shortness of breath)–10 (maximal shortness of breath) | 23 | 2.63 ± 1.23 | 19 | 2.64 ± 1.89 | 0.494 |
| FEV1/FVC | >70–80% | 19 | 54.11 ± 17.41 | 19 | 63.47 ± 14.23 | 0.039 |
| COPD CAT score | 5 minimal impact, and >30 very high impact | 25 | 21.2 ± 8.46 | 22 | 20.64 ± 6.73 | 0.401 |
| $SaO_2$ | 95–100%[26] | 19 | 91.32 ± 4.88 | 19 | 91.32 ± 5.95 | 1.000 |
| GAD-7 score | Refer to Table 1 | 86 | 5.70 ± 5.92 | 85 | 4.52 ± 4.90 | 0.157 |
| PHQ-9 score | Refer to Table 1 | 86 | 7.03 ± 6.16 | 85 | 6.16 ± 4.62 | 0.285 |
| AQoL-8D score | Range 0–1 (full health) | 86 | 0.58 ± 0.22 | 85 | 0.58 ± 0.21 | 0.872 |

SD: standard deviation; COPD: chronic obstructive pulmonary disease; PRaDA: Pattern Recognition and Data Analytics; FEV1: Forced Expiratory Volume; FVC: Forced Vital Capacity; COPD CAT: COPD Assessment Test; GAD-7: Generalized Anxiety Disorder 7-item; PHQ-9: Patient Health Questionnaire 9-item; AQoL-8D: Assessment of Quality of Life.
*Median.
**People with diabetes.
***Percent of population with microalbuminuria.

The difference of 0.09 over 12 months between the two groups was statistically significant (95% CI: 0.05, 0.14) in favour of the intervention group (Table 3).

*Secondary outcomes*

1. Clinical outcomes

Health-related clinical data did not show a statistically significant difference between groups at 12 months (Table 3), with the exception of urine albumin–creatinine ratios in the population with diabetes.

2. Anxiety and depression

There was no statistical difference in the mean GAD-7 and PHQ-9 scores between the intervention and control groups at baseline. At 12 months both the GAD-7 and PHQ-9 scores had improved in the intervention group, whereas the mean scores for both instruments in the

**Table 3.** Primary and secondary outcomes.

| Outcome | Normal values | *n* | Study groups 12 months Intervention | *n* | Control | Difference between groups 95% CI Intervention–control | *p*-value |
|---|---|---|---|---|---|---|---|
| Primary outcomes | | | | | | | |
| Total number of hospital admissions (*n*) | N/A | 86 | 102 | 85 | 130 | –28 | |
| Proportion of participants who had at least one hospital admission | N/A | 46 | 53% | 47 | 55% | −2% | 0.813 |
| Mean number of hospital admissions (all patients) | N/A | 86 | 1.19 ± 1.56 | 85 | 1.53 ± 2.00 | −0.34 (–0.19, 0.88) | 0.106 |
| Mean number of hospital admissions (admitted patients only) | N/A | 46 | 2.22 ± 1.50 | 47 | 2.77 ± 1.96 | −0.55 (–1.27, 0.17) | 0.067 |
| Acute admissions | N/A | 86 | 1.13 ± 1.48 | 85 | 1.47 ± 1.91 | −0.34 (–0.17, 0.85) | 0.095 |
| Subacute admissions | N/A | 86 | 0.06 ± 0.24 | 85 | 0.06 ± 0.24 | 0.01 (–0.07, 0.07) | 0.493 |
| Mean hospital LOS | N/A | 86 | 5.69 ± 16.36 | 85 | 9.58 ± 19.97 | −3.89 (–9.40, 1.62) | 0.083 |
| Acute admissions | N/A | 86 | 4.56 ± 11.71 | 85 | 8.66 ± 17.35 | −4.1 (−8.56, −0.36) | 0.036** |
| Subacute admissions | N/A | 86 | 1.13 ± 5.20 | 85 | 0.92 ± 4.02 | −0.21 (−1.61, 1.19) | 0.616 |
| Mean hospital cost (AUD$) | N/A | 86 | 5510 ± 13,508 | 85 | 12,063 ± 22,482 | −6553 (−12,145, −961) | 0.011** |
| Acute admissions (AUD$) | N/A | 86 | 4651 ± 10,085 | 85 | 11,271 ± 20,705 | −6620 (−11,528, −1712) | 0.004** |
| AQoL-8D score | N/A | 83 | 0.63 ± 0.03 | 80 | 0.54 ± 0.03 | 0.09 (0.05, 0.14) | ** |
| Secondary outcomes | | | | | | | |
| Blood pressure | <80 mmHg | 66 | 130.60 ± 5.18 | 51 | 132.10 ± 5.85 | −1.48 (−9.29, 6.33) | |
| Diastolic | <120 mmHg | | 75.11 ± 3.41 | | 73.70 ± 3.87 | 1.42 (−3.72, 6.57) | |
| Systolic | | | | | | | |
| HbA1c | 6.5–7% | 51 | 5.76 ± 0.45 | 47 | 5.97 ± 0.46 | −0.22 (−0.87, 0.43) | |
| Total cholesterol | <5.5 mmol/l | 50 | 3.98 ± 1.17 | 44 | 4.93 ± 1.23 | −0.94 (−2.65, 0.76) | |
| HDL cholesterol | 0.9–2 mmol/l | 49 | 1.14 ± 0.07 | 43 | 1.15 ± 0.07 | −0.01 (−0.12, 0.09) | |
| LDL cholesterol | 2–3.4 mmol/l | 48 | 1.88 ± 0.21 | 42 | 1.88 ± 0.22 | −0.02 (−0.30, 0.30) | |
| Triglyceride | <1.7 mmol/l | 50 | 2.08 ± 0.19 | 44 | 2.04 ± 0.20 | −0.04 (−0.23, 0.32) | |
| Microalbumin | <2.5 mg/mmol | 43 | 7.29 ± 6.89 | 34 | 18.69 ± 8.47 | −11.40 (−22.32, −0.48) | ** |
| Proportion* | | 37% | | 62% | | | |
| 6-min walk test distance | 348–371 m[25] | 13 | 272.99 ± 46.96 | **7** | 270.1 ± 64.37 | 2.80 (−76.29, 81.90) | |
| Borg scale (dyspnoea) | Modified Borg scale: 0–10 | 12 | 2.30 ± 0.69 | 6 | 1.44 ± 0.97 | 0.85 (−0.35, 2.05) | |
| FEV1/FVC | >70% normal | 14 | 57.79 ± 15.84 | 8 | 64.00 ± 19.53 | 6.21(−9.71, 22.14) | |
| COPD CAT score | 5 minimal impact, and >30 very high impact | 15 | 19.06 ± 0.86 | 13 | 18.87 ± 0.92 | 0.75 (−0.51, 1.45) | |
| $SaO_2$ | 95–100%[26] | 10 | 92.20 ± 2.21 | 7 | 93.64 ± 2.96 | −1.44 (−5.13, 2.25) | |
| GAD-7 score | | 83 | 3.00 ± 0.87 | 80 | 5.45 ± 0.88 | −2.46 (−3.69, −1.20) | ** |
| PHQ-9 score | | 83 | 4.85 ± 1.02 | 80 | 8.32 ± 1.04 | −3.46 (−4.91, −2.01) | ** |

CI: confidence interval; LOS: length of stay; AQoL-8D: Assessment of Quality of Life; FEV1: Forced Expiratory Volume; FVC: Forced Vital Capacity; COPD CAT: Chronic Obstructive Pulmonary Disease (COPD) Assessment Test; GAD-7: Generalized Anxiety Disorder 7-item; PHQ-9: Patient Health Questionnaire 9-item;

*Proportion of participants with microalbuminuria (>2.5mg/mmol).

**Statistically significant at 0.05.

**Table 4.** Health education impact at 12 months.

| | Intervention $n = 69$ | | Control $n = 59$ | | | 95% CI | | |
|---|---|---|---|---|---|---|---|---|
| HeiQ domains | MD* | SD | MD* | SD | MD** | Lower | Upper | *p*-value |
| Health-directed behaviour | 0.09 | 0.49 | −0.08 | 0.69 | −0.17 | −0.37 | 0.04 | 0.113 |
| Positive and active engagement in life | 0.11 | 0.59 | −0.16 | 0.55 | −0.27 | −0.47 | −0.07 | 0.010 |
| Self-monitoring and insight | 0.13 | 0.44 | −0.08 | 0.43 | −0.21 | −0.36 | −0.06 | 0.008 |
| Constructive attitudes and approaches | 0.12 | 0.45 | −0.01 | 0.48 | −0.13 | −0.29 | 0.03 | 0.120 |
| Skill and technique acquisition | 0.17 | 0.52 | −0.05 | 0.50 | −0.22 | −0.40 | −0.04 | 0.018 |
| Social integration and support | 0.12 | 0.52 | −0.19 | 0.63 | −0.30 | −0.50 | −0.10 | 0.018 |
| Health services navigation | −0.10 | 0.75 | 0.02 | 0.57 | 0.12 | −0.12 | 0.36 | 0.313 |
| Emotional distress | 0.04 | 0.45 | 0.00 | 0.55 | −0.05 | −0.22 | 0.13 | 0.594 |

CI: confidence interval; SD: standard deviation.
*Mean difference two between time points.
**Mean difference between groups

control group worsened. The difference at 12 months for both scores showed a statistically significant change in favour of the intervention group (Table 3), suggesting lower anxiety and depression in the intervention group compared to the control group at 12 months.

3. Health literacy

There was no difference in the mean Health Education Impact Questionnaire (HeiQ) score between study groups at baseline. However, at 12 months, four domains (positive and active engagement in life; self-monitoring and insight; skill and technique acquisition; and social integration and support) out of eight domains showed a statistically significant difference between the two time points (Table 4).

At 12 months there were six domains (health-directed behaviour; positive and active engagement in life; self-monitoring and insight; constructive attitudes and approaches; social integration and support; and emotional distress) that showed an improvement in the intervention group. In the control group, there was an improvement in only one domain (health services navigation).

4. Cost-effectiveness

When the costs of the telehealth intervention were included in addition to the costs of hospitalisation, there was no statistically significant difference in the mean total cost between groups (intervention: AUD$12,796 versus control: AUD$12,081; MD AUD$714; 95% CI: −4879, 6308). The ICER based on the incremental cost per quality-adjusted life year (QALY) between intervention and control groups was a cost of AUD$7933 per QALY saved.

## Discussion

The Barwon Health personalised telehealth monitoring program showed a significant difference in hospital LOS and hospital cost between study groups favourable to the remote patient monitoring intervention group with a mean saving of 3.9 bed days. Although there was no significant difference in the number of hospital admissions between the two groups, from the hospital's perspective a difference of 28 admissions in 12 months is relevant in terms of freeing up beds for other patients, or savings costs. Much of the research investigating telehealth monitoring effectiveness has reported favourable outcomes[10,11] in terms of acute and subacute hospitalisations. However, the cost-minimisation study by Paré et al.[12] of a telehealth monitoring program for people with COPD reported that the intervention group had fewer hospital admissions, but the LOS was longer than for the control group. These results are different to our study with the savings in the costs of hospitalisation due to both a reduction in mean length of stay over 12 months as well the total number of admissions.

Although there are many studies of telehealth monitoring interventions, there is limited evidence about the economic benefits. A systematic review conducted by Cruz et al.[13] reported a saving of AUD$6553 (95% CI: –12145, –961) per person in the costs of hospitalisation over 12 months. This saving is consistent with our results. Similar to our study, others have also reported that the costs of telehealth monitoring are high,[14] so the savings in hospitalisation are likely to offset the higher costs of the intervention. Although the difference in total costs between the two groups was not statistically significant in our pilot study of a small sample of patients, we believe that over time, as the technology evolves, so the costs will decrease. Due to the savings in LOS there are potential economic benefits related to improved patient flow that generate additional income for acute services. There are also opportunities for economies of scale at the hospital level by increasing the number of patients who are eligible and willing to participate in the telehealth monitoring program.

Cartwright et al.[15] suggested that detailed professional attention and self-monitoring can result in better disease

management, improve patient's health literacy, reduce anxiety and depression, and reduce healthcare visits. These results are consistent with our findings with improvement observed in HRQOL, anxiety, depression and health literacy. Due to a close relationship with clinicians and better disease management, we might expect improvement in patient's quality of life. The difference in the AQoL-8D utility index of 0.09 is likely to be clinically significant and is equivalent to an annual gain of one month of full health. Health literacy refers to the degree to which individuals have the capacity to obtain, process and understand basic health information,[16] to understand their disease, their treatments and the alternatives to improve their health status. This includes understanding their need for different health services and the ability to make decisions that improve their health and well-being.[17] Improvement in functional health literacy is likely to improve the individual's ability to access appropriate healthcare services, manage medication and understand health information provided to them.[17]

The progressive increase of albumin in urine is one of the characteristics of diabetic kidney disease[18] and may be predictive of end-stage renal disease.[19] Better control of glycaemia and arterial blood pressure may reduce the incidence of microalbuminuria and end-stage renal failure.[20] In our study, the intervention group showed significant improvement in microalbumin levels in urine between baseline and 12 months. Since there was no major change in blood pressure or haemoglobin A1c between baseline and 12 months, we believe that this improvement may be due to the combined effect of a slight lowering of blood pressure and slightly better glucose control in the intervention group.

The cost-effectiveness analysis showed an ICER of AUD$7933 per QALY saved based on the incremental difference in costs and HRQOL between the intervention and control groups. This means that the cost of a gain of one year in full health would be AUD$7933. Whilst there is no publicly stated willingness to pay threshold for a QALY in Australia, it has routinely been assumed that AUD$50,000 per QALY is an acceptable threshold in evaluating economic evidence for the listing of new pharmaceuticals on the Pharmaceutical Benefits Scheme. The ICER of AUD$7933/QALY for the intervention compared to usual care is below a willingness to pay threshold of AUD$50,000/QALY, suggesting that the Barwon Health Personalized Health Care Project intervention is value for money.

The major strengths of this research are the inclusion of the actual intervention costs, including the personalised telehealth remote patient monitoring set-up and labour costs according to the actual staff delivering the program and the time taken. Secondly, data included many clinical indicators at baseline and 12 months relevant to diabetes and/or COPD. Our study had some limitations. Firstly, it was a pilot design with a small sample not powered to determine a significant difference in the primary outcomes of costs or HRQOL. Secondly, we were not able to collect data on community-based services such as GP visits and other health service utilisation that was not part of the health service provider, Barwon Health. Inadvertently, additional patients were randomised to either intervention or control groups who did not meet the original inclusion criteria (PRaDA score); however, if results were affected, they are likely to be a conservative estimate, as the mean PRaDA score at baseline for the control group was lower than the intervention group, suggesting a lower probability of hospital readmission in the control group.

## *Conclusion*

The Barwon Health personalised telehealth remote patient monitoring intervention was associated with savings in hospitalisation that offset the higher costs of the telehealth intervention. The improvement in HRQOL, anxiety, depression and health literacy observed in the intervention group is likely to be clinically relevant. The ICER of AUD$7933/QALY saved is below the AUD$50,000/QALY threshold that is generally accepted in an Australian policy context.

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## Supplementary material

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